

AHA SCIENTIFIC STATEMENT

Multimodality Approach to Coronary Ischemic Testing in Pediatric Patients: A Scientific Statement From the American Heart Association

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ABSTRACT: As new research reveals the value of provocative and ischemic coronary testing in various acquired and congenital coronary issues in children, an unmet need for targeted and thoughtful provocative ischemic testing emerges as a high priority for our field. A number of evidence-based guidelines and scientific statements for conditions including Kawasaki disease, multisystem inflammatory syndrome in children associated with coronary aneurysms, anomalous aortic origin of the coronary arteries, cardiac allograft vasculopathy, and surgically repaired congenital heart disease involving coronary artery manipulation now recommend provocative ischemic testing in children. The strengths and limitations of various testing modalities and provocative agents are explored, including the evidence for use in children. In addition, evidence-based adaptations and recommendations for optimizing use of these technologies in children are noted. The pediatric cardiac imaging community requires more information to understand appropriate use of existing provocative ischemic modalities and to help develop the technology for future use.

Key Words: AHA scientific statements ■ anomalous coronary artery ■ cardiac imaging techniques ■ congenital heart defects ■ coronary artery disease ■ mucocutaneous lymph node syndrome

The main objective of provocative ischemic testing is to assess the functional or physiologic significance of anatomic coronary artery abnormalities. There is a growing experience with provocative ischemic testing in children for a variety of coronary artery abnormalities (both congenital and acquired), as new research has shown feasibility, appropriateness, and evolving utility for such testing. Although several evidence-based guidelines and scientific statements exist for the use of provocative ischemic testing in adults with epicardial coronary artery disease,¹ existing evidence in pediatrics is mainly limited to single-center studies and pediatric-specific guidance is lacking. There is an unmet need for targeted and thoughtful provocative ischemic testing strategies for a variety of

pediatric cardiac diseases, including Kawasaki disease, multisystem inflammatory syndrome in children associated with coronary aneurysms, anomalous aortic origin of the coronary arteries (AAOCA), cardiac allograft vasculopathy (CAV), and surgically repaired congenital heart disease involving coronary artery manipulation.^{2,3}

When assessing for provocative ischemia in children, additional physiologic and technical aspects must be considered for a successful test. Both the physiology of the stressor (exercise versus pharmacologic) and the age and developmental stage of the child have effects on tolerance and ability to participate. Some ischemic testing methods require intravenous access, exposure to ionizing radiation, compliance with exercise protocols, or fitting of cumbersome equipment,

which limit their feasibility in children. The spatial and temporal resolution of each test may also limit diagnostic efficacy in children with small hearts and high heart rates. Finally, the need for sedation or general anesthesia to complete testing must be considered in the overall risk/benefit ratio of the test or tests, when multimodality testing strategies are required.

Due to the high variability of symptoms from coronary insufficiency in children, there is a tendency to use provocative testing for screening before onset of symptoms in high-risk populations, as compared with adult cardiology where provocative testing is mainly used to work up symptoms of angina in well-defined intermediate and high-risk groups. However, the assessment of ischemic risk in pediatric coronary disease is hampered by lack of data, often relying on expert consensus. As the use of provocative ischemic testing for pediatric coronary disease becomes more common, a review of the current literature focusing on critical differences in physiology, pathophysiology, technical performance, and diagnostic efficacy from adults, and disease-specific risk-stratification and value in directing management/surgical intervention in children may be helpful.

CAUSE, PATHOPHYSIOLOGY, AND APPROACHES TO TESTING FOR MYOCARDIAL ISCHEMIA

The Ischemic Cascade and Provocative Agents

Provocative ischemic testing involves increasing myocardial oxygen demand, coronary blood flow by means of coronary dilation, or both, to identify areas with insufficient coronary blood flow. The ischemic cascade describes the sequence of events that occur due to insufficient coronary blood flow, beginning with metabolic changes, followed by wall motion abnormalities (WMA), ECG changes, and ending with angina and myocardial infarction, [Figure 1](#).⁴ This sequential progression of the ischemic cascade is supported by the fact that myocardial perfusion abnormalities are more common than WMA during dobutamine stress cardiovascular magnetic resonance (CMR) imaging in adults with coronary artery stenosis and pediatric patients with AAOCA.^{5,6}

Provocative agents vary in their mechanism of action. It is important to understand their strengths and limitations for a given pathology when choosing the appropriate test, particularly for children of varying ages and developmental stages. Differences in these agents at rest and during stress represent the capacity of coronary flow to increase during hyperemia.⁷

Adenosine, dipyridamole, and regadenoson are coronary vasodilators, used as hyperemic agents for stress testing in fixed coronary obstructions.⁸ They do not simulate exercise physiology and are not a good choice for dynamic coronary obstruction. Adenosine and dipyridamole are nonselective adenosine receptor agonists, whereas regadenoson is selective for A_{2A}, resulting in fewer noncardiac side effects.^{9,10} Adenosine and dipyridamole are administered as infusions, requiring 2 intravenous lines, one for the agent and another for contrast, whereas regadenoson is given as a bolus, followed by contrast administration, therefore requiring only a single intravenous line. This can be an important consideration for imaging in younger patients without sedation.

Dobutamine is a beta-1-agonist, reproducing some features of exercise by increasing oxygen demand via increased heart rate and a significant increase in myocardial contractility.^{6,11} It has also been shown to provide maximal coronary vasodilation at high dose.¹¹ Dobutamine is less physiologic than exercise stress, lacking the augmentation of preload that comes with exercise. Whereas end-diastolic volume increases with exercise stress echocardiography (ESE) due to increased systemic venous return and preload, dobutamine stress echocardiography (DSE) results in a decreased end-diastolic volume and absence of stroke volume augmentation.¹² It is the preferred stressor agent in patients with potential dynamic narrowing of the coronary artery⁶ or those who cannot tolerate vasodilators (eg, asthma). Prior literature reports low sensitivity of DSE imaging to detect WMA. Nevertheless, adding first pass perfusion defect detection with dobutamine stress enhances the sensitivity to detect ischemia.^{5,6} Atropine may be given in addition to dobutamine infusion to achieve target heart rates. It is important to monitor for arrhythmias and ischemia.

Exercise is considered the reference standard for stress testing, as it preserves the normal electromechanical, metabolic, and physiological response. Exercise stress should be used for ischemic testing if possible, to allow for symptom correlation. Maximal exercise results in significant increases in heart rate, systolic blood pressure, stroke volume, and mild increases in contractility and myocardial blood flow. A variety of exercise protocols exist for both treadmill and cycle ergometer testing, creating provocative options for children each with an appropriate determination of a maximal versus submaximal test.¹³ Not all testing modalities are conducive to using exercise. For example, exercise stress CMR has been studied¹⁴ although the setup is not widely available and patients are required to lie flat and minimize motion, which often limits exercise capacity. Emerging

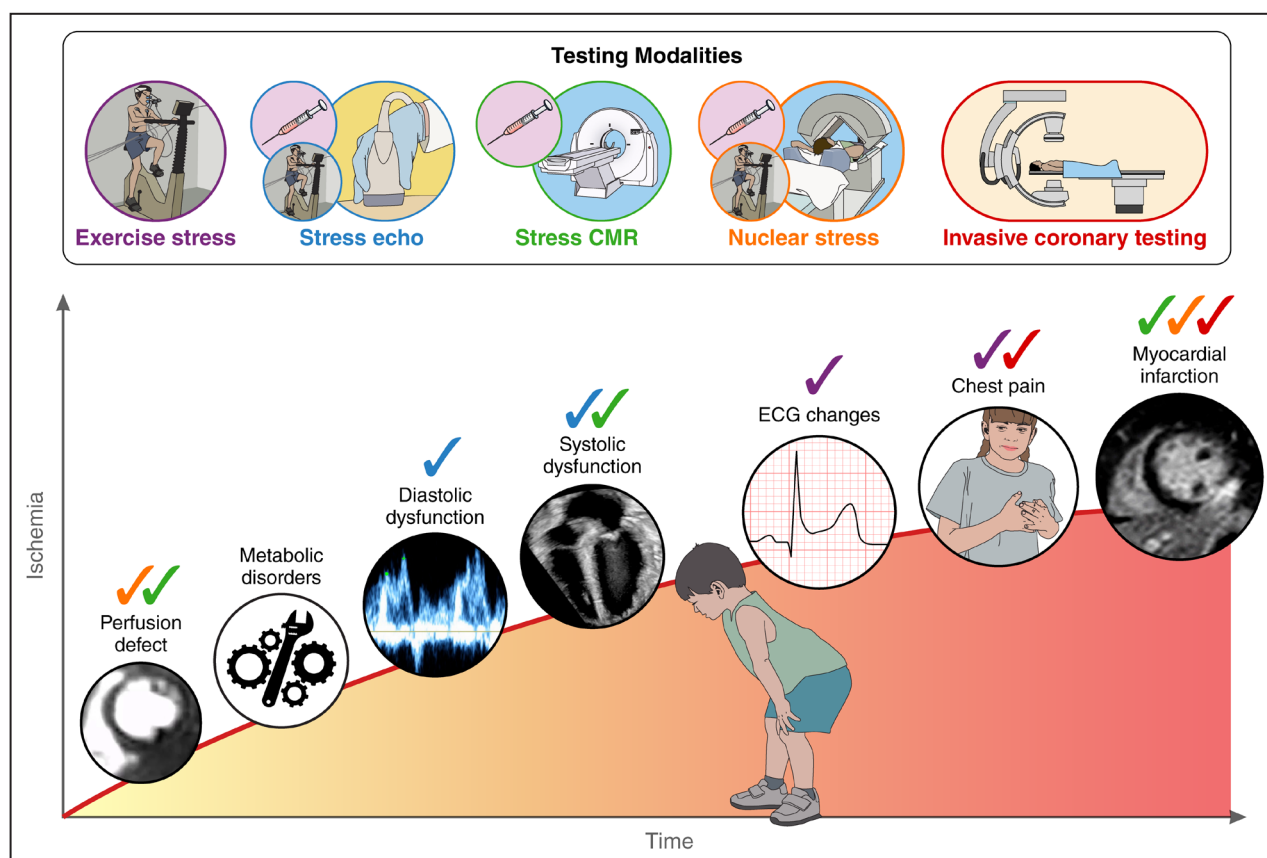


Figure 1. The ischemic cascade in children. Changes are apparent with different testing modalities at different time points in the ischemic cascade, thus each test has different sensitive, specificity, and appropriateness depending on the clinical situation. CMR indicates cardiac magnetic resonance.

research suggests the potential to combine submaximal exercise with low-dose pharmacologic agents to improve diagnostic yield.¹⁵

It is important to note that, regardless of provocation type, all patients should be monitored closely for ischemia and arrhythmia by properly licensed and trained professionals for detection of abnormalities and have clear criteria for intervention and cessation of the test.

DISEASE-SPECIFIC MECHANISMS OF ISCHEMIA

Pediatric coronary conditions create ischemia in a multitude of ways (Table 1). In Kawasaki disease (KD), ischemia is provoked by epicardial coronary inflammation, thrombosis, and vessel remodeling, which creates fixed obstruction. A similar situation arises in CAV in heart transplant patients and can coexist with microvascular obstruction to cause ischemia. In surgically manipulated coronaries in congenital heart disease, coronary lesions tend to be ostial or very proximal within the coronary bed, again creating fixed obstruction of the majority of the coronary territory. AAOCA

can create ischemia by blending both fixed and dynamic obstruction in the coronary bed. Myocardial bridges create a purer dynamic obstruction but can coexist with hypertrophic cardiomyopathy, which can be subject to microvascular obstruction. Finally, acute myopericarditis can frequently present with signs and symptoms mimicking acute coronary syndrome and thus often an anatomic coronary evaluation may be initiated in these patients too.

TYPICAL ISCHEMIC TESTING IN ADULT PATIENTS

The 2021 American College of Cardiology/American Heart Association Chest Pain guidelines detail various risk assessment algorithms and clinical decision pathways that classify patients as low, intermediate, or high risk based on the clinical history, ECG, and cardiac biomarker values and determine downstream evaluation. High-risk patients with acute chest pain, who are identified as experiencing acute coronary syndrome, are triaged directly to invasive coronary angiography. In low-risk patients, it is acceptable to pursue no further

Table 1. Evidence Supporting Ischemic Testing by the Type of Diagnosis

Coronary artery condition	Variations	Physiology for ischemia	Provocative ischemia test	Pediatric age considerations	Considerations
AAOCA	Wide anatomic variation: standardized anatomic assessment with coronary computed tomography angiography (including vessel, sinus of origin, proximal course, orifice shape, etc) should be performed before considering provocative ischemic testing modality	Dynamic obstruction, coexisting fixed obstruction	ESE DSE	Must consider age and size for type of exercise in ESE Rapid return to baseline HR after exercise <8 y old may need general anesthesia for DSE	Exercise the most physiologic stress in AAOCA Lack of published data correlating outcomes with positive provocative ischemic testing in AAOCA Lack of standardization of exercise protocols in AAOCA
			Nuclear perfusion	Significant radiation dose Mainly performed at adult centers, which may have minimum age requirements May be performed at outpatient centers (vs the main hospital)	Can use exercise or pharmacologic stress Extensive experience with this modality Prone to false positives, particularly in older female patients
			Dobutamine stress CMR	<12 y old may require general anesthesia Non-sedated patients may have issues with tolerance	Majority of single-center published data reported on DSCMR Target for adequate stress still being studied Long-term follow-up and outcome data needed
			Invasive coronary assessment	More invasive test Requires additional equipment that is not standard in pediatric laboratories for IVUS and FFR/IFR Less technical expertise for IVUS/FFR/IFR in smaller patients	Absence of FFR/IFR cutoffs in this anatomy Safety data in pediatric patients is needed Lack of data using dobutamine vs adenosine for FFR calculation IFR offers the advantage of not requiring additional medications
Inflammatory coronary disease	Coronary artery aneurysms with thrombosis/stenosis/occlusion as sequelae from Kawasaki disease, multisystem inflammatory syndrome in children, other vasculitides	Fixed stenosis	ESE DSE	Young patients	Some literature for DSE and ESE in identifying coronary artery stenosis Long-term follow-up and outcome data needed
			Adenosine stress CMR	Young patients (<1 y old) are highest risk group for aneurysms, and to develop thrombosis: will likely need general anesthesia Often causes hypotension in younger patients under GA Image resolution, particularly during perfusion imaging, may be a concern for small children	Regadenoson and adenosine are safe and very well tolerated Can assess for occult myocardial infarction with delayed enhancement imaging Coronary collateralization may preserve measurements of perfusion in the setting of stenoses Some literature supports dobutamine stress CMR
			Invasive coronary assessment	More invasive test Less technical expertise in smaller patients Requires additional equipment that is not standard in pediatric laboratories for IVUS, optical coherence tomography, and FFR/IFR	Best characterization of vessel wall, including abnormal remodeling, and collateral formation Remains reference standard for identifying coronary stenosis Limited data on FFR cutoffs in coronary aneurysms

(Continued)

Table 1. Continued

Coronary artery condition	Variations	Physiology for ischemia	Provocative ischemia test	Pediatric age considerations	Considerations
Postsurgical	Procedures involving coronary manipulation, eg, dextro-loop transposition of the great arteries status post arterial switch operation, Ross operation	Fixed stenosis	Nuclear perfusion	Significant radiation dose Mainly performed at adult centers, which may have age limits May be performed at outpatient centers (vs the main hospital)	Can use exercise or pharmacologic stress Extensive experience with this modality Presence of devices/prosthetic material/conduits can impact interpretation
			Adenosine stress CMR	<8 y old may need general anesthesia Often causes hypotension in younger patients under general anesthesia	Regadenoson and adenosine are safe and very well tolerated Some literature in postsurgical population showing utility in identifying reimplanted coronary stenosis/occlusion Utility of screening vs clinically driven testing needs to be determined
			Invasive coronary assessment	More invasive test May be only coronary testing option for postoperative patients with ventricular dysfunction on extracorporeal membrane oxygenation	Remains reference standard for identifying coronary stenosis
			ESE DSE	Must consider postoperative precautions, age, and size for type of exercise in ESE Rapid return to baseline HR after exercise <8 y old may need general anesthesia for DSE	Some literature for DSE and ESE in identifying coronary artery stenosis Long-term follow-up and outcome data needed
CAV	Epicardial disease, microvascular dysfunction	Fixed stenosis	Adenosine stress CMR		Majority of literature in adult transplant patients Pediatric literature shows it is safe despite concerns for conduction abnormalities Quantitative perfusion may allow for detection of microvascular dysfunction
			Invasive coronary assessment	Reference standard Frequent arterial percutaneous access needed Requires additional equipment that is not standard in pediatric laboratories for IVUS and FFR/IFR	Interoperator variability in diagnostic grading of CAV IVUS and FFR may provide useful adjunctive information in assessing CAV
Myocardial bridge	Deep (>2 mm), symptomatic	Dynamic obstruction	ESE DSE	Must consider age and size for type of exercise in ESE <8 y old may need general anesthesia for DSE	Majority of literature on evaluation of myocardial bridges
			Dobutamine stress CMR	<12 y old may require general anesthesia Non-sedated patients may have issues with tolerance	Some literature showing good agreement between perfusion abnormalities in DSCMR and abnormal FFR/IFR
			Invasive catheterization	More invasive test Requires additional equipment that is not standard in pediatric laboratories for IVUS and FFR/IFR	Limited data on ideal FFR/IFR cutoffs Potential for vasospasm

AAOCA indicates anomalous aortic origin of the coronary artery; CAV, coronary allograft vasculopathy; CMR, cardiac magnetic resonance; DSCMR, dobutamine stress cardiac magnetic resonance; DSE, dobutamine stress echocardiography; ESE, exercise stress echocardiography; FFR, fractional flow reserve; HR, heart rate; IFR, instantaneous wave free ratio; and IVUS, intravascular ultrasound.

testing or defer testing to an outpatient setting. The overall approach can be classified into either an anatomic testing strategy, namely coronary computed tomography angiography, or a functional testing strategy, involving stress testing.

Diagnostic strategies need to have good discrimination to identify those at highest risk and avoid unnecessary downstream testing in those at lowest risk. The choice of diagnostic strategy is ideally a patient-centric decision, determined by individual risk profile, demographic factors, specific clinical questions, and local resources. The identification of obstructive coronary artery disease (>50% stenosis) on coronary computed tomography angiography, along with evidence of lesion-specific ischemia, or the presence of moderate to severe ischemia on functional testing, identifies those who would benefit from invasive coronary angiography.

A PEDIATRIC APPROACH TO ISCHEMIC TESTING

More often than in adults, there is substantial local variation of available first-line diagnostic modalities, as well as expertise for ischemia assessment in children. The same patient population may undergo different stress testing at different institutions. This underscores the importance of standardizing diagnostic approaches in pediatric ischemia testing. Common indications for ischemic testing in children include disease surveillance in high-risk individuals with predisposing conditions, workup for exertional symptoms concerning for ischemia in any individual, and testing to help clarify anatomic or other testing suggesting ischemia (ie, abnormal ECG or echocardiogram).

Although ischemia is rare in the pediatric population, there are subsets of patients with increased risk of myocardial ischemia and sudden cardiac death, as highlighted in [Table 1](#). KD with a history of current or regressed aneurysms is a leading cause of acquired coronary artery disease in infants and children, present in $\approx 0.03\%$ of children in the United States. Even with timely treatment, severe inflammation of the coronary arteries can lead to aneurysm formation, stasis of blood, subsequent coronary thrombosis, and eventual coronary stenosis.² Acquired pediatric coronary disease also includes history of heart transplantation with transplant-vasculopathy, and history of chest radiation. Genetic or congenital causes include AAOCA, myocardial bridging, surgically repaired congenital heart disease involving coronary artery manipulation, pulmonary atresia with intact ventricular septum and coronary fistulae, and familial homozygous hyperlipidemia. Congenital heart defects such as dextro-loop transposition of the great arteries and AAOCA are present in

≈ 1 in 4000 and at least 1 to 10 in 1000 newborns; repair of these conditions can increase risk of ischemia, in part due to the long-term risk of developing coronary ostial stenosis post repair.³ With the growing popularity of the Ross procedure for aortic valve disease, it is important to monitor the coronary arteries of this population in the same way.

Functional testing is underused and less frequently performed in pediatrics compared with anatomic testing and should be given more consideration in evaluating children for ischemia. Ideally, functional ischemic testing would be combined with anatomic characterization of the coronary tree to determine dominance pattern to understand if anatomic targets are present for intervention, particularly in dextro-loop transposition of the great arteries and I-looped ventricles following switch operation.

A systematic, patient-centric approach to coronary and provocative testing determined by individual risk profile is needed but several challenges bear mentioning. One of the challenges is that ischemia in children can be difficult to suspect and identify. Children may have difficulty in articulating symptoms, and symptoms can be incorrectly ascribed to non-cardiac causes without further investigation secondary to the rarity of ischemia in children. Furthermore, children with ischemia who have received chest radiation or those who have had heart transplantation may not experience chest pain with ischemia due to alteration of the typical innervation of the heart. Many infants and children also develop significant collateralization despite significant occlusion or flow abnormalities, such as in anomalous origin of the left coronary artery from the pulmonary artery, and may be completely asymptomatic. Due to the myriad causes, it is challenging to reliably classify patients as low, intermediate, or high risk, with the majority of children falling into a low-risk category. With the growth of multicenter registries and larger sample sizes, improved risk stratification may be on the horizon.

Decision-Making for Treatment and Interventions for Positive Ischemic Tests in Children

The goals of testing for ischemia include identification of presence, severity, and management decision-making regarding need, candidacy, and type of intervention. Interventional options for ischemia are strongly based on the cause, whether in the adult or pediatric population. Some options are identical in both populations, such as unroofing for myocardial bridges, and the spectrum of surgical options for AAOCA repair.¹⁶ Children are not small adults, and risk assessment and management strategies differ significantly in different

age and size groups. For example, percutaneous coronary intervention including balloon +/- stent angioplasty, and coronary artery bypass grafting for surgical revascularization are primarily used in adults. Both strategies are generally avoided in children whenever possible due to the technical limitations imposed by their smaller size and the growth potential and longevity required in children.

In children with KD, there is frequently the need to use the internal mammary arteries for surgical coronary artery bypass grafting when indicated, because venous conduits do not perform as well due to poor growth potential and increased incidence of thrombosis.¹⁷ The durability of grafts is an important consideration because children have a longer life than the duration of their bypass grafts.¹⁸ Postoperative complications include reoperation due to vessel stenosis/kinking, thrombosis, new ischemia, new aortic valve insufficiency, and new ventricular dysfunction; these may be more significant in children than in adults and play an important role in shared decision-making for a given condition.¹⁹

OVERVIEW OF PROVOCATIVE ISCHEMIC TESTING MODALITIES

There are many different types of provocative ischemic testing, although not all modalities may be available at a given institution. Table 1 reviews common pediatric coronary artery conditions, and Table S1 reviews the suggested testing modality based on the pathophysiology of ischemia and currently available literature. A brief review of the various modalities is included here, along with special considerations for optimizing testing in children.

Modality 1: Exercise Stress Testing

Cardiopulmonary stress testing (CPET) is an essential tool to evaluate cardiovascular and pulmonary function in pediatric patients. This test assesses functional capacity, oxygen use, and myocardial ischemia by employing treadmill or bicycle ergometry.²⁰ CPET is designed to detect the early stages of the ischemia cascade, including flow irregularities and subclinical myocardial dysfunction, before overt symptoms like chest pain arise (Figure 1). CPET assesses exercise tolerance and detects subclinical pulmonary or cardiovascular impairment, offering critical insights into disease severity and guiding therapeutic interventions.²¹

Sensitivity for detecting myocardial ischemia using CPET combined with stress imaging (eg, echocardiography or nuclear imaging) in children ranges from 75% to 85%, whereas standalone CPET has a sensitivity of ~50% to 60%, as it relies on indirect markers like ST-segment changes, which are less common in

pediatric ischemia.²¹ Specificity for ischemia detection is generally higher, ranging from 80% to 90%, particularly when stress imaging is included to differentiate nonischemic causes of abnormalities.^{21,22}

The choice between treadmill and bicycle ergometry varies based on patient characteristics (height should be >135 cm) and diagnostic goals. Treadmills mimic natural activity, making them suitable for children with preserved mobility and coordination. However, they may cause motion artifacts in both ECG and imaging-based tests. Bicycles allow for better imaging quality due to reduced motion but may underestimate exercise capacity because cycling is less familiar to most children and works fewer muscle groups. Ruiz et al. reported that supine bicycle ergometry provides superior imaging quality while still enabling functional assessments.²³

Modality 2: Stress Echocardiography in Children

Mechanism

Stress echocardiography allows for the dynamic assessment of coronary flow reserve and regional wall motion abnormalities. Stress echocardiography is a fundamental test for assessment of ischemia in adults, with a sensitivity of 72% to 85% and specificity of 80% to 95%.²⁴ Recently, its feasibility and long-term prognostic utility have also been demonstrated in pediatric populations.²⁵ Both exercise and pharmacologic stress can be used, which increases its versatility. At-rest assessment in both exercise and pharmacologic stress echocardiography should incorporate multimodal echo imaging to improve diagnostic yield. A comprehensive assessment of a child should include 2-dimensional and 3-dimensional imaging, strain imaging, and evaluation of diastolic function including Doppler acquisition.²⁶ For the left ventricle, the standard 17-segment model is widely adopted, whereas a 9-segment model has been proposed for systematic right ventricular analysis, allowing more detailed evaluation of regional function.¹² Given clinical indications and the limited time window for post-exercise imaging, a tailored subset of these options may be included to offer higher sensitivity for subclinical myocardial dysfunction and ischemia.

Exercise Stress Echocardiography

The strengths of ESE are the absence of needing intravenous access, contrast dye, or radiation exposure, its portability, and its cost-effectiveness. Therefore, it can be broadly used for detection of ischemia in high-risk patients and in long-term serial follow-up of patients who have undergone surgeries/interventions for ischemia. Furthermore, ESE allows the physician to

evaluate the patient's functional capacity in a physiological state that most closely mimics that of an active child. The onset of new WMA occurs before symptoms or ECG changes (Figure 1), which improves sensitivity compared with exercise testing alone and is the basis of ischemia detection and stress echo interpretation.

Pharmacologic Stress Echocardiography

In children who cannot exercise or are too young to perform a treadmill or bicycle study (typically <6–10 years), pharmacological stress, usually with dobutamine, is a useful alternative and can be performed under general anesthesia in young infants and children. DSE requires more monitoring than ESE, and patient tolerance is an important consideration for nonsedated children. Higher doses are associated with headache, chest discomfort, nausea, and occasionally vomiting, but with coaching and rapid image acquisition, children as young as 7 years can complete the test without sedation. Due to its portable nature, pharmacological stress can be advantageous in intensive care settings and in evaluation of postoperative myopathic issues. Additionally, contrast-enhanced echocardiography is a valuable adjunct in patients with suboptimal image quality or suspected subendocardial ischemia. The use of contrast agents can improve endocardial border delineation in both ESE and pharmacological stress echocardiography and enable real-time perfusion assessment.²⁷

Indications

The utility of stress echocardiography has been demonstrated in many pediatric populations at high risk for ischemia. In patients with KD, who may have coronary aneurysms that progress to coronary artery disease, DSE demonstrated excellent specificity and negative predictive value, with wall motion score index providing independent prognostic value for major adverse cardiac events at 15-year follow-up.²⁸ ESE is also increasingly used as a noninvasive way to assess myocardial dysfunction. Tedla et al. reviewed their experience with ESE in patients with KD with coronary aneurysms and found coronary aneurysms, stenosis, and myocardial perfusion defects in all patients with abnormal ESE (n=3), highlighting the utility of ESE as a screening tool for high-risk patients with KD.²² Similarly, Linglart et al. reviewed their institutional experience with patients who underwent arterial switch operation for D-looped transposition of the great arteries.²⁹ They reported a significant incidence of stenosis-related events, emphasizing the need for early diagnosis and intervention. Parallels with patients who have any surgical manipulation of their coronaries (Ross procedure, AAOCA, etc) may provide insights for managing these patients with abnormal coronary anatomy. In cardiac

transplantation patients, a positive (or abnormal) stress echo for ischemia was predictive of higher risk of myocardial infarction, death, or need for retransplantation, as compared with those with a negative stress echo, regardless of findings on angiography.²⁵

Limitations

Limitations to stress echocardiography interpretation are typically encountered in those with difficult acoustic windows or prior history of myocardial infarction, in whom detection of ischemia may be less sensitive. Additionally, patients who are unable to exercise can undergo DSE but may require sedation. It is also important to have appropriate equipment, personnel, and protocols in place. Another important consideration is proper training in the acquisition and maintenance of skills in stress echocardiography interpretation, which are vital to the detection of ischemia. Ideally stress echo sonographers would have experience in both pediatric and adult settings, where opportunities to learn stress echo imaging are more plentiful. Assessment of WMA is not a standard skill among pediatric cardiac imagers.

Protocols

A basic overview of ESE and DSE protocols is highlighted in Table 2.³⁰ For exercise protocols, a treadmill is preferred, especially in younger children, given its ability to elicit higher myocardial oxygen demands. There tends to be a rapid heart rate recovery in children post exercise, which increases the importance of completing postexercise imaging in 60 to 90 seconds. Rapid heart rate recovery, however, is also a good prognostic sign because it indicates significant cardiac reserve, and therefore, generally less ischemia. When DSE is performed under general anesthesia, as often necessary in young children, there is already before the test a lowering of blood pressure and heart rate and thus a greater difficulty in achieving maximal heart rate and blood pressure response.

Modality 3: Nuclear Stress Imaging in Children

Nuclear perfusion imaging involves administration of a radiopharmaceutical with affinity for myocardium whose uptake depends on coronary flow and cell viability to demonstrate areas of decreased perfusion (Figure 2). Imaging is obtained using single-photon emission computed tomography (SPECT) or positron emission tomography (PET), with use driven by availability at any given institution. SPECT with exercise or pharmacological stress is currently more widely available. In children, choices of SPECT agents include ^{99m}Tc-Sestamibi and ^{99m}Tc-Tetrofosmin, while [¹⁵O]H₂O, [¹³N]NH₃, ⁸²Rb and [¹⁸F]Flurpiridaz are agents that may be used for positron

Table 2. Practical Considerations for Ischemic and Coronary Artery Testing by Modality, With Special Considerations for Children

	Typical protocol	Considerations in children
Exercise stress echocardiography	Treadmill or bicycle (upright or supine) exercise using Bruce or ramp (preferred) protocol. Resistance increased by 20–25 W at 3 min intervals until maximal target HR is reached or test terminated because of fatigue or symptoms. Images obtained immediately post exercise.	Post-exercise imaging should be completed in 60–90 sec due to rapid HR recovery in children. In children, there may be lower success in eliciting maximal HR with bicycle protocols as compared with treadmill. Bicycle protocols may only be feasible in taller children who can reach the pedals.
Dobutamine stress echocardiography	Dobutamine delivered at 3-min intervals, typically 10, 20, 30, 40 mcg/kg/min. Atropine administered if ~85% of the maximal HR for age is not reached. Imaging performed prestress and with each stage as indicated. Dobutamine at low doses (5, 10, 20 mcg/kg/min) can help identify myocardial viability or cardiac contractile reserve.	Dobutamine infusion can be increased to 50 mg/kg/min in children to achieve HR and blood pressure targets. General anesthesia typically used with dobutamine stress echocardiography may lead to inability to achieve 85% maximal HR and blood pressure goal. Atropine delivered to children at a low dose (0.01 mg/kg). Preparation includes holding beta-blockers, avoiding caffeine.
Dobutamine stress MRI	Identical dobutamine administration parameters and atropine considerations. T1-weighted perfusion sequence acquired during dobutamine infusion. Patients require 2 intravenous lines.	Similar issues with general anesthesia. Matrix and field of view optimized to capture perfusion defects in smaller hearts. Preparation includes holding beta-blockers, avoiding caffeine.
Adenosine stress MRI	Adenosine infusion 140 mcg/kg/min for 3 min (patients require 2 intravenous lines). Regadenoson administration with aminophylline reversal (patients require 1 intravenous line).	Similar issues with general anesthesia. Same issues with spatial and temporal resolution.
Nuclear perfusion	Similar adenosine, regadenoson and dobutamine parameters as previously. Exercise treadmill testing feasibility varies by institution with SPECT vs PET.	Spatial resolution superior with PET than SPECT. Radiation exposure less with PET than SPECT.
Invasive coronary assessment	Before advancing the intracoronary imaging device (OCT or IVUS wires), secure a therapeutic range of ACT with proper IV heparin titration. Administer intracoronary nitroglycerine (dose of 83.9 ± 44.3 mcg/kg for right coronary artery and 93.0 ± 46.7 mcg/kg for left coronary artery). Manual contrast injection (as opposed to power injection) is preferred, especially in young patients avoids the risk of dissection (contrast injection is required for OCT, but not for intravascular ultrasound imaging).	Repeat ACT monitoring, IV heparin, and nitroglycerin as needed according to time laps of the procedure. Secure the proper guiding catheter in the coronary ostia with the possibility of intracoronary pressure monitoring.

ACT indicates activated clotting time; HR, heart rate; MRI, magnetic resonance imaging; OCT, optical coherence tomography; PET, positron emission tomography; and SPECT, single photon emission computed tomography.

emission tomography perfusion.^{31,32} Spatial resolution in SPECT is enhanced using converging collimators in children.³³ Positron emission tomography offers higher spatial resolution than SPECT with potential for quantitating absolute perfusion and myocardial flow reserve but has traditionally been limited by need for an on-site generator or cyclotron. 18F-Flurpiridaz has recently been approved by the FDA for myocardial perfusion imaging for coronary artery disease in adults, and has a 110-minute half-life, offering the ability to be produced at a regional cyclotron facility and shipped to the hospital as needed, and may be especially cost-effective for low-frequency pediatric use.³⁴ Choice of stressor is determined by age and disease, with the advantage of being able to use both exercise and pharmacologic stressors. While

treadmill or cycle ergometry is used in older children, especially in AAOCA, the use of pharmacological stress with regadenoson or adenosine is preferred in smaller children, KD, and after coronary manipulation. Coronary microvascular disease can coexist with epicardial coronary disease in post-transplant cardiomyopathy, KD, hypertrophic cardiomyopathy and certain dilated cardiomyopathies,³⁵ and may be assessed using quantitative myocardial blood flow at rest/maximal hyperemia and coronary vasodilatory capacity (ratio of hyperemic MBF to resting MBF). There is no data on coronary flow reserve in children, and this represents an unexploited area for future research.³⁶ When compared with CMR, the main limitation with nuclear stress perfusion is exposure to radiation, which is 3.7 mSv (3.2–4.1) with

positron emission tomography, and substantially higher with SPECT of 12.8 mSv (12.2–14.3).³⁷ It is the modality of choice to assess myocardial perfusion in children based on diagnostic and safety considerations³³ except in 2 situations where nuclear testing may be preferred: (1) CMR compatibility or safety issues prevent the performance of CMR, and (2) when conventional treadmill stress is desirable as with AAOCA.

Modality 4: Stress Cardiovascular Magnetic Resonance in Children

Technique

Stress CMR involves induction of myocardial hyperemia and increasing heart rate using pharmacological stress agents followed by intravenous administration of a gadolinium-based contrast agent during a rapid T1-weighted perfusion sequence to assess serial transit through the heart and myocardial tissue, allowing for regional assessment of perfusion at stress and rest. Stress CMR has emerging feasibility with exercise at this time but is limited by the need for specialized equipment and training, and thus pharmacologic provocation is the main method used. The main stressor agents used in children are adenosine, regadenoson, and dobutamine, with safety and efficacy of a weight-based protocol in children shown in several studies.^{6,10,38} Each stress agent requires some variations in protocol, as shown in [Table 1](#).

Clinical Applications

Recent literature has shown the utility of stress CMR in a variety of coronary artery conditions. Stress CMR is a valuable adjunct in the functional assessment of patients with AAOCA, in whom CPET does not provide definitive findings congruent with ischemia, likely associated with the low sensitivity of the test for ischemia in this condition.³⁹ Dobutamine stress CMR has been shown to be safe and feasible in pediatric patients with AAOCA, and it has demonstrated the resolution of perfusion abnormalities following successful surgical repairs.^{6,10} Following coronary artery surgery for AAOCA, it is imperative to ensure that it has indeed been resolved and the success of the surgery has been validated. Surprisingly, the Congenital Heart Surgeons' Society AAOCA registry reports that procedural success was not always confirmed with provocative ischemic testing, with a higher than expected proportion of patients demonstrating new ischemia.¹⁹ First-line postoperative investigations include ECG, echocardiography, computed tomographic angiography and CPET for clearance,⁴⁰ and stress CMR can provide functional myocardial perfusion data and anatomic data. Of note, it is crucial to document the presence of preoperative scar using

CMR late gadolinium enhancement, to have a baseline for comparison.

The pathophysiology of coronary artery stenosis in KD with aneurysms is complex, posing management challenges.⁴¹ Stress perfusion CMR with vasodilators is increasingly used to detect myocardial perfusion defect, WMA, and scarring, aiding in risk stratification and identifying subclinical ischemia.^{10,42,43} Detection of CAV in orthotopic heart transplant by stress perfusion CMR in adult patients has variable sensitivity (41%–100%) and specificity (61%–100%) when compared with invasive coronary angiography and intravascular ultrasound (IVUS).⁴⁴ Single-center studies have demonstrated feasibility and safety of regadenoson stress CMR in pediatric orthotopic heart transplant patients but sensitivity/specificity for detecting CAV has not yet been reported.⁴⁵ As quantitative myocardial perfusion sequences become more readily available, additional studies on its utility with KD and CAV may show improved clinical prognostic ability. Quantitative perfusion imaging may capture coronary collateralization flow in the presence of significant stenosis.

Strengths

The overall strengths of stress CMR lie in its ability to detect the ischemic cascade at multiple points. It offers comprehensive noninvasive assessment of segmental perfusion, function, wall motion, viability, and parametric mapping in a single study. For myocardial perfusion, CMR has better spatial resolution and fewer artifacts than nuclear perfusion and has been extensively validated in adult coronary artery disease for diagnosis, prognosis, risk stratification, and guidance of coronary interventions.⁴⁶ Finally, CMR can detect both epicardial coronary disease and microvascular dysfunction. Stress CMR requires the administration of intravenous gadolinium, an intravenous agent that causes T1 relaxivity and allows for perfusion and fibrosis imaging.

Limitations

Challenges in children include need for sedation to overcome motion/discomfort related to the intravenous lines and pharmacological stress, fast heart rates, respiratory motion, complex anatomy, higher incidence of dynamic coronary ischemia, and different mechanisms of myocardium and coronary vessels to adapt to ischemia.³³ CMR with stress perfusion can have limited anatomical detail in smaller children due to patient size and high heart rates. Young children may require sedation, and provocative perfusion imaging in this situation has not been extensively studied. Another consideration is CMR use in patients with

implantable devices, which can be limited by imaging artifacts, MR compatibility, and safety issues. The CMR technique in children can be abbreviated by using free-breathing real-time sequences, but this is not widely available at this time. Although research supports its utility, current evidence is limited by small sample sizes from single-center studies. And similar to stress echocardiography, assessment of perfusion defects and myocardial viability, although common in adult cardiovascular imaging, is not a foundation of pediatric CMR training. Future studies should focus on standardizing techniques and management strategy using multicenter collaboration.

Modality 5: Invasive Coronary Testing in Children

Invasive coronary angiography (CAG) remains essential for coronary assessment in children. Selective CAG with intracoronary injection of iodine-based radiopaque dye provides instantaneous high-resolution imaging. Recently, IVUS coronary assessment has been introduced in children to minimize iodine contrast side effects and provide additional information regarding the vessel wall structure.⁴⁷ IVUS provides in-lesion measurement, stent selection, and deployment for optimal apposition results and surveillance for secondary dissection.⁴⁸ IVUS image quality is limited by the low resolution as it is based on ultrasound echo-processing. Competing advantages were brought in by optical coherence tomography (OCT) using near-infrared interferometric technology.⁴⁹ OCT provides a much higher axial resolution (10–20 μm) and better 3-dimensional renderings. Despite lesser tissue penetration, OCT has diagnostic potential in KD and other lesions with coronary stenosis.⁵⁰

Fractional flow reserve (FFR) and instantaneous wave-free ratio (iFR) are intracoronary physiological indices used to assess the severity of intermediate coronary artery stenosis primarily in adults, but with increasing relevance in pediatrics.^{51,52} Both techniques measure the pressure ratio across a lesion with an intravascular wire, comparing distal coronary pressure to aortic pressure. Whereas FFR is measured during induced hyperemia (adenosine), considering the mean pressure across the entire cardiac cycle, iFR is measured at baseline during diastole only, when coronary resistance is naturally minimized. An FFR ≤ 0.80 or iFR ≤ 0.89 indicates significant stenosis that may benefit from revascularization in adults.^{51,53}

In pediatric patients with fixed coronary artery lesions, FFR and iFR can provide valuable information for management.^{52,54} In coronary anomalies with dynamic obstruction as the potential mechanism of ischemia, dobutamine provocation is preferred, and diastolic FFR or iFR is used to avoid a falsely normal FFR from overshooting of coronary systolic

pressure.^{6,8,55} Invasive assessment of AAOCA and myocardial bridging includes coronary angiography, FFR/iFR (dobutamine), and IVUS or OCT.^{8,52,55–57} In intramural/interarterial AAOCA, FFR/iFR can be considered in challenging cases^{55,58}, and IVUS/OCT have been used to demonstrate vessel wall compression in adults.⁵⁹ In myocardial bridges or intraseptal AAOCA, FFR/iFR help quantify ischemia and IVUS/OCT assess systolic compression.⁶⁰ These techniques can guide clinical decisions, but the risks must be discussed with the families, and they must be performed by trained experts. Invasive imaging in KD should be limited to patients with clinically significant coronary artery aneurysms.² CAG provides a complete image of the coronary tree, can detect and quantify stenosis, blood flow characteristics, and coronary artery aneurysms.²² CAG is particularly useful in very young patients with fast heart rates that can complicate noninvasive, gated imaging modalities. FFR, IVUS, and OCT can provide data on the changes of the coronary artery wall and guide future treatment.⁵⁴ CAG is frequently used in patients following orthotopic transplant to monitor for CAV. CAG is highly specific in the posttransplantation population but only moderately sensitive for detection of CAV.⁶¹ More recently, IVUS has been used in children down to 10 kg, with greater sensitivity for CAV in patients ≥ 5 years post transplant versus CAG.⁴⁵ The use of FFR and coronary flow reserve provides important diagnostic and prognostic information by assessing coronary physiology.⁶² Additionally, OCT has potential to detect coronary changes earlier than CAG, allowing for earlier intervention and improved outcomes.⁶³

Limitations of invasive angiography include risk of in situ injury and must be weighed against therapeutic benefit. These techniques are invasive and provocative, and thus carry risk, are less validated in children, and are not performed widely. They should be performed only by physicians trained to do so in these populations. Other limitations include ionizing radiation, a concern shared with coronary computed tomography angiography, osmotic renal injury, and iodine allergy.⁶⁴

FUTURE DIRECTIONS

Due to increases in the prevalence of pediatric coronary disease and guideline-driven testing, there is an urgent need to develop and study imaging technology and normative provocative testing data for more precise assessment of coronary disease and ischemia in children. Understanding the individual test sensitivity and specificity for a given population, and therefore the implications of a positive or negative provocative test in a child, are paramount to clinical decision-making. Adult cardiac testing benefits from a codified risk stratification score, which does not exist in pediatric cardiology; multidisciplinary efforts to develop a

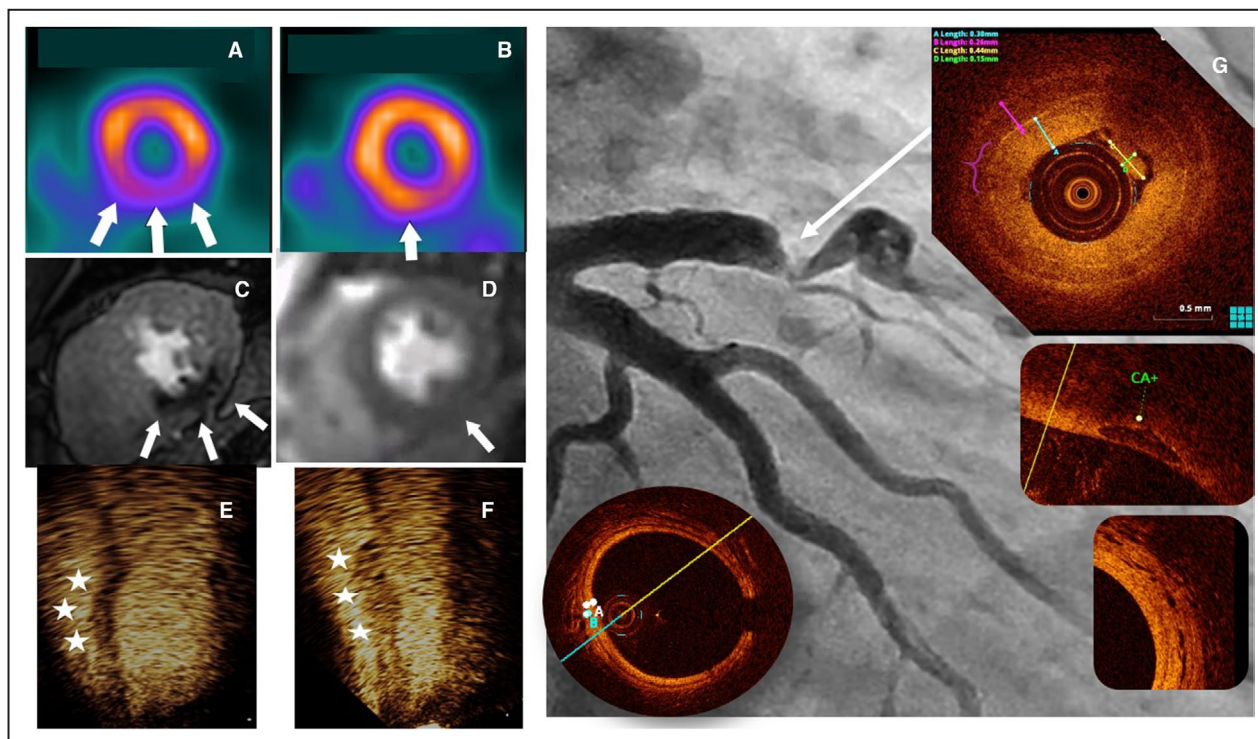


Figure 2. Imaging modalities that identify coronary disease

A, B, Stress and rest nuclear testing, respectively, demonstrating an inferior perfusion defect (3 arrows) that reverses (1 arrow) with rest. **C, D,** Stress and rest cardiac magnetic resonance imaging testing respectively, demonstrating an inferior perfusion defect (3 arrows) that reverses (1 arrow) with rest. **E, F,** Stress and rest contrast-enhanced dobutamine stress echocardiography respectively, demonstrating an anterosseptal regional wall motion abnormality. **G,** Single plan invasive coronary angiography with severe LAD stenosis (1 arrow) and accompanying OCT images of the coronary bed. Normal coronary OCT image (bottom left insert) with 3 well-differentiated layers, (internal to external) the intima/endothelium (bright orange) and the media/smooth muscle layer (subsequent dark layer) are thin and homogeneous, followed by the adventitia (external layer). This is contrasted in the same patient with pathological findings (top right insert) at the site of progressive stenotic lesion where a past giant coronary aneurysm had regressed in size at the mid-LAD segment (arrow). The stenotic segment demonstrates neointimal hyperplastic proliferation (blue calipers: 300 μm), hyperplastic media layer (pink calipers: 260 μm), with irregular border fibrotic inclusions (bracket) and a red clot with its cast shadow effect (1 to 3 o'clock) between the catheter and the neointima. A calcific plaque deposit (bottom middle insert) has the typically well-defined circumference and a faint color (void of hemoglobin) that, unlike hemoglobin-rich clots, does not cause a shadow effect as these lesions are translucent (unlike the calcium shadow caused by ultrasound imaging). Finally, neovascularization can be seen typically at the periphery of the injured segments (bottom right insert). LAD indicates left anterior descending coronary artery; and OCT, optical coherence tomography.

risk scoring system for children are needed to help standardize testing indications. Testing interpretation stands to improve both in quality and in standardization through involvement of artificial intelligence in image analysis and interpretation of wall motion scoring, subtle anatomic features, and changes in function.⁶⁵ Furthermore, in the future, greater familiarity, education, and development of pediatric stress imagers will further improve the accessibility, sensitivity, specificity, and utility of these tests in children.

Pediatric ischemic imaging must be treated as a distinct skill requiring its own training; thus cardiology and radiology fellowships should include exposure to provocative testing, including positive testing, which is difficult due to the rarity of disease at any given center. It is important that pediatric provocative imaging programs work with adult imaging experts to gain

more clinical exposure to understanding regional wall motion analysis, invasive angiography, and emerging imaging technologies. Congenital programs must invest in new technology, such as higher resolution FFR with OCT, and expertise to understand how to discern acquired and congenital coronary artery issues in babies and children. Given the rarity of outcomes needed for validating new diagnostic approaches in children, development of multi-institutional registries and use of physical and computational approaches for studying ischemia in pediatric conditions for development of predictive models must be explored. The pediatric cardiac imaging community requires more systematic, multisite information to understand the strengths, limitations, and applications of existing provocative ischemic modalities, and to help develop the technology for future use.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

This statement was approved by the American Heart Association Science Advisory and Coordinating Committee on August 18, 2025, and the American Heart Association Executive Committee on December 8, 2025. A copy of the document is available at <https://professional.heart.org/statements> by using either "Search for Guidelines & Statements" or the "Browse by Topic" area. Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/JAHA.125.047790>

The American Heart Association requests that this document be cited as follows: Olivieri LJ, Lee S, Chen MH, Dahdah N, Doan TT, Jegatheeswaran A, Krishnamurthy R, Tierney ESS, Weber EM, Rao A; on behalf of the American Heart Association Council on Lifelong Congenital Heart Disease and Heart Health in the Young and the Council on Cardiovascular and Stroke Nursing. Multimodality Approach to Coronary Ischemic Testing in Pediatric Patients: A Scientific Statement From the American Heart Association. *J Am Heart Assoc.* 2026;15:e047790. doi: [10.1161/JAHA.125.047790](https://doi.org/10.1161/JAHA.125.047790)

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Disclosures

Writing Group Disclosures

Writing group member	Employment	Research grant	Other research support	Speakers' bureau/honoraria	Expert witness	Ownership interest	Consultant/advisory board	Other
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This table represents the relationships of writing group members that may be perceived as actual or reasonably perceived conflicts of interest as reported on the Disclosure Questionnaire, which all members of the writing group are required to complete and submit. A relationship is considered to be "significant" if (a) the person receives \$5 000 or more during any 12-month period, or 5% or more of the person's gross income; or (b) the person owns 5% or more of the voting stock or share of the entity, or owns \$5 000 or more of the fair market value of the entity. A relationship is considered to be "modest" if it is less than "significant" under the preceding definition.

*Modest.

Reviewer Disclosures

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This table represents the relationships of reviewers that may be perceived as actual or reasonably perceived conflicts of interest as reported on the Disclosure Questionnaire, which all reviewers are required to complete and submit. A relationship is considered to be "significant" if (1) the individual receives $\geq \$5000$ during any 12-month period, or $\geq 5\%$ of the individual's gross income; or (2) the individual owns $\geq 5\%$ of the voting stock or share of the entity, or owns $\geq \$5000$ or more of the fair market value of the entity. A relationship is considered to be "modest" if it is less than "significant" under the preceding definition.

*Modest.

†Significant.

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